SMARTBRIDGE
Let's Bridge the Gap

Project Initialization and Planning Phase

Date	15 March 2024
Team ID	
Project Title	Health360: Executive Analytics & Population Health Dashboard
Maximum Marks	3 Marks

Project Proposal (Proposed Solution) template

This project proposal outlines a solution to address a specific problem. With a clear objective, defined scope, and a concise problem statement, the proposed solution details the approach, key features, and resource requirements, including hardware, software, and personnel.

Project Overview	
Objective	To design, build, and publish an interactive, single-screen Tableau Master Dashboard and a 9-point guided Executive Tableau Story that transforms raw healthcare data into actionable clinical, behavioral, and demographic insights for healthcare leaders.
Scope	Includes data processing, calculated measure creation (Age Category, Calculated Risk Score), visual analytics across 9 dedicated worksheets, dynamic global cross-filtering, and cloud deployment on Tableau Public Server for a 5,000-patient synthetic cohort.
Problem Statement	
Description	Healthcare decision-makers face fragmented, static, and tabular patient datasets that obscure vital correlations between clinical metrics (Blood Pressure, Cholesterol), lifestyle behaviors (Smoking, Alcohol, Exercise, BMI), and overall population health risks across different life stages.
Impact	Solving this problem prevents delayed clinical interventions, enables targeted preventive wellness spending, optimizes facility resource planning, and speeds up executive decision-making across all age brackets (Children to Elder).

Proposed Solution	
Approach	Utilize an agile, single-contributor Business Intelligence workflow: clean and transform raw tabular patient data, engineer custom calculated fields, construct focused visual charts, combine them into an interactive single-page Master Dashboard, and compose a 9-point guided Tableau Story for executive presentation.
Key Features	• Pre-calculated dynamic metrics (Age Category, Calculated Risk Score). • Clinical vitals scatter plot correlating Blood Pressure, Cholesterol, and Risk. • Multi-factor behavioural risk heatmap (Smoking, Alcohol, Physical Activity). • Demographic life-stage stratification (Children to Elder) and symptom tracking. • Real-time global cross-filtering across all 9 connected dashboard worksheets.

Resource Requirements

Resource Type	Description	Specification/Allocation
Hardware		
Computing Resources	Standard CPU for data processing and dashboard rendering	Quad-Core Intel Core i5/i7 or Apple M-Series processor
Memory	System RAM for流畅 Tableau Desktop extract processing	8 GB RAM (16 GB Recommended)
Storage	Disk space for raw dataset, Tableau workbook, and extracts	500 MB free SSD storage space
Software		
Frameworks / Tools	Business Intelligence and Data Visualization Platform	Tableau Desktop Professional & Tableau Public Server
Libraries / Query Logic	Calculated Syntax Engine for business rules & logic	Native Tableau Calculated Fields & Table Calculations

Development Environment	Visual workspace and file versioning	Tableau Desktop Workspace (.twbx format)
Data		
Data	Flat-file structured healthcare dataset	smart_healthcare_dataset.csv (5,000 patient records)